SETTING UP A LOCAL AREA NETWORK (LAN)

EXP-4

AIM:-

To design setup and configure a local area network (LAN) connecting four computers through a network switch & demonstrate resource / data sharing across the network.

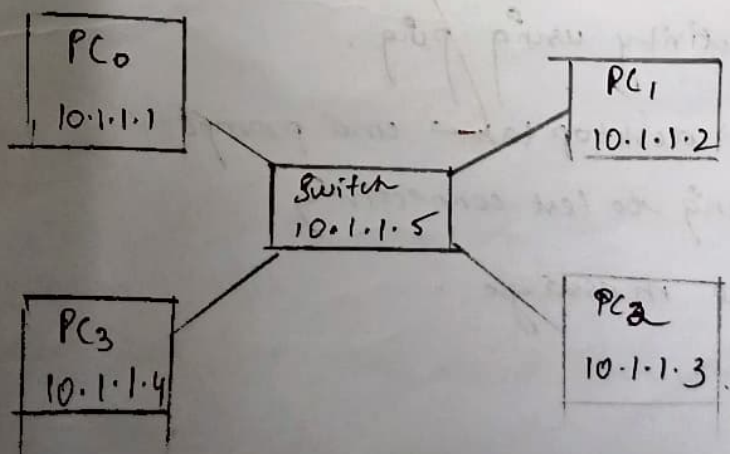
EQUIPMENT REQUIRED:-

Equipment	Qty	Specification
Personal Computer	4	Ethernet (NIC) enabled)
Network switch	1	8/16/24 Port
Ethernet cables (cat5e / cat6)	4	Straight through, RJ 45 connectors.

IP Address configuration tables:-

Devices	IP address	Subnet mask	Connected to
PC_0	10.1.1.1	255.0.0.0	switch Port Fa0/1
PC_1	10.1.1.2	255.0.0.0	switch Port Fa0/2
PC_2	10.1.1.3	255.0.0.0	switch Port Fa0/3
PC_3	10.1.1.4	255.0.0.0	switch Port Fa0/4
Pswitch	10.1.1.5	255.0.0.0	VLAN interface.

NETWORK DIAGRAM:-



Implementation in Cisco Packet Tracers

1) Place the devices

- open Cisco Packet Tracers
- from the device category panel, select end devices & drag 4 PCs onto the workspace
- select switches & drag one switch.

2) Cable the devices

- click connections → select copper straight-through cable
- click each PC then click the switch & select an available FAST ethernet port to complete each connection.
- Repeat for all four PCs (Port Fa0/1, to Fa0/4). Green link lights confirm a physical successful condition

3) Assign IP addresses to PC's

- click a PC → Desktop tab → IP configuration
- Enter the static IP address and subnet mask as per the IP table.

4) Configure the switch management IP

- click the switch → CLI tab
- enable
  configure terminal - interface VLAN-1 - ip address 10.1.1.5. 255.0.0.0 - no shut down - exit.

5) Verify connectivity using ping.

- click a PC → desktop tab → cmd prompt.
- use the ping to test connectivity
- successful message.

6) Demonstrate Data sharing (server simulation)

- Add a server device (End devices → server-PT) and connect it to the switch
- Assign it an IP address
- Open the server's services tab & enable the HTTP
- From another PC, open web browser and enter the server's IP address to load the page confirming successful data transfer across the LAN.

RESULT:-

Thus LAN set tled up & connection is verified successfully.